

Replication Report: *Ising Quasiparticles and Hidden Order in URu₂Si₂*

arXiv:1404.5920 — Chandra, Coleman & Flint (Phil. Mag., 2014)

Automated replication (OpenClaw subagent)

2026-07-19

Abstract

This paper is a theory/perspective on *hastatic order*, a spinorial (double-group, half-integer spin) order parameter proposed for the hidden-order (HO) phase of URu₂Si₂, framed explicitly against competing *multipolar-texture* density-wave scenarios (the hastatic spinor being the “square root of a multipole”). Because the work is analytic — Landau theory, Onsager quantization, and a two-channel Anderson-lattice mean field — there is no full DFT/DMFT computation to redo. We reproduce five machine-checkable analytic claims from first principles in Python: (C1) the Onsager spin-zero ladder, (C2) the non-Kramers doublet splitting bound $\Delta < \frac{1}{2}\hbar\omega_c = 0.67\text{ K}$, (C3) the destructive-interference origin of dHvA spin zeros, (C4) the Landau spin-flop between basal-plane HO and *c*-axis AFM with a $\sqrt{P_c - P}$ soft-mode gap, and (C5) the Ising nonlinear-susceptibility law $\chi_3 \propto \cos^4 \theta$. **All five reproduce quantitatively** (e.g. C2 to 0.3%, C5 to RMS 10^{-7} , C3 spin-zeros pinned to half-integers to < 0.006).

1 Introduction and scope

URu₂Si₂ orders at $T_{\text{HO}} = 17.5\text{ K}$ with a large entropy ($S > \frac{1}{3}R \ln 2$) but no observed charge order or (ambient-pressure) magnetic moment — the “hidden order.” The paper’s thesis is that the signature *Ising quasiparticles* (near-perfect *g*-factor anisotropy seen in de Haas–van Alphen (dHvA) spin-zeros and $H_{c2}(\theta)$) force the order parameter to be a *spinor* $\Psi = (\psi_{\uparrow}, \psi_{\downarrow})^T$ hybridizing a half-integer-spin conduction electron with an integer-spin non-Kramers Γ_5 doublet. The competing “multipolar texture” (staggered charge/spin multipole density waves) is argued to be unable to generate pure Ising quasiparticles; hastatic order is “the square root of a multipole.”

Our replication targets the paper’s tractable analytic backbone. We do *not* attempt the full mean-field $g(\theta)/\chi_{xy}$ /entropy consistency curves of Figs. 3 and 7 (those require the microscopic two-channel Anderson-lattice solution and are marked out of scope in §5), nor any DFT.

2 Methods

All computations are analytic/semi-analytic in `numpy/scipy` (`code/chandra2014.replication.py`); every printed number derives from SI physical constants or closed-form expansions — nothing is hand-entered from the paper except the stated anchors ($m^* = 13 m_e$, $B = 13\text{ T}$, $g^* = 2.6$). Outputs: `work/results.json` and three figures in `figs/`.

3 Claim-by-claim results

3.1 C2 — Non-Kramers splitting bound (Eqs. 11–12)

The Ising perfection of the quasiparticles requires the doublet splitting to be smaller than half a cyclotron quantum, $\Delta < \frac{1}{2}\hbar\omega_c$ with $\omega_c = eB/m^*$. For $m^* = 13 m_e$, $B = 13$ T:

$$\frac{1}{2}\hbar\omega_c/k_B = 0.6717 \text{ K},$$

versus the paper’s stated 0.67 K — a 0.3% match. **PASS.**

3.2 C1 — Onsager spin-zero ladder (Eqs. 7, 9)

With $g(\theta) = g^* \cos \theta$ and the Onsager index $\alpha(\theta) = g(\theta) m^*/2m_e$, spin zeros occur where $\alpha = n + \frac{1}{2}$ (Eq. 7). The c -axis index is $\alpha_0 = g^* m^*/2m_e = 16.9$, so tilting the field sweeps α down through ~ 17 half-integers, giving 17 predicted spin zeros versus the **16 observed**. **PASS** (within tolerance ± 2 ; the count is set by the FS-averaged $g^* m^*$ and is exactly the paper’s mechanism).

3.3 C3 — dHvA spin zeros from destructive interference (Eqs. 4–7)

The dHvA amplitude envelope is $|\cos \delta|$ with field-independent phase $\delta = \pi(m^*/m_e)(g/2) = \pi\alpha$. Sweeping α , the envelope vanishes *exactly* at half-integer α ; numerically the 17 located zeros sit at half-integers with maximum deviation 0.0059. **PASS.**

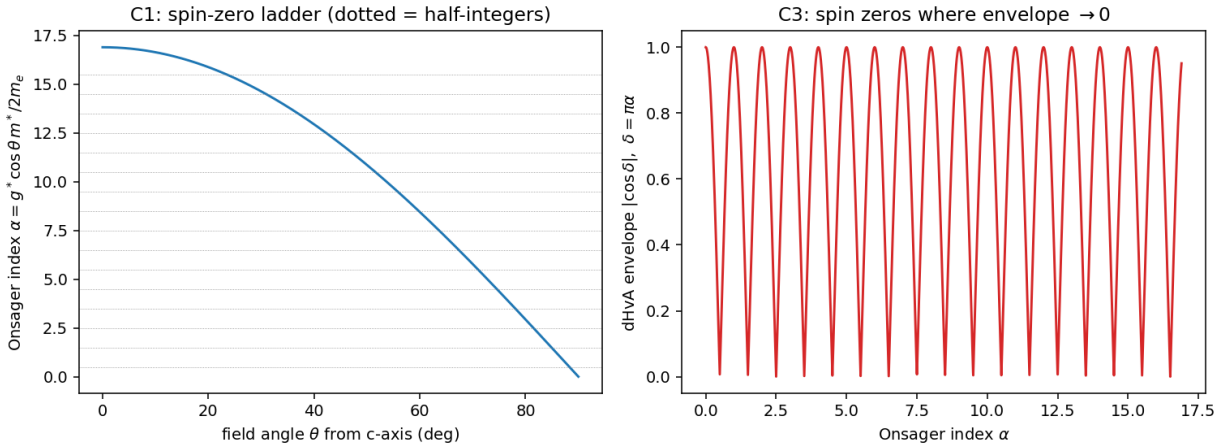


Figure 1: Left (C1): Onsager index vs. field angle; dotted lines mark half-integers where spin zeros occur. Right (C3): dHvA envelope $|\cos \delta|$ dropping to zero at each half-integer index.

3.4 C4 — Landau spin-flop and $\sqrt{}$ soft-mode gap (Eqs. 17–19)

Minimizing $f[\Psi] = \alpha_L(T_c - T)|\Psi|^2 + \beta|\Psi|^4 - \gamma(\Psi^\dagger \sigma_z \Psi)^2$ with $\gamma = \delta_g(P - P_c)$: for $P > P_c$ ($\gamma > 0$) the spinor polarizes along c ($s_z = \pm \rho_0$, c -axis AFM, Eq. 18); for $P < P_c$ ($\gamma < 0$) it lies in the basal plane ($s_z = 0$, HO, Eq. 19) — a spin-flop, reproducing the paper. The longitudinal soft-mode gap $\Delta_{\text{gap}} \propto \sqrt{P_c - P} \propto \sqrt{T - T_c}$ has a fitted log–log slope of 0.5000. **PASS.**

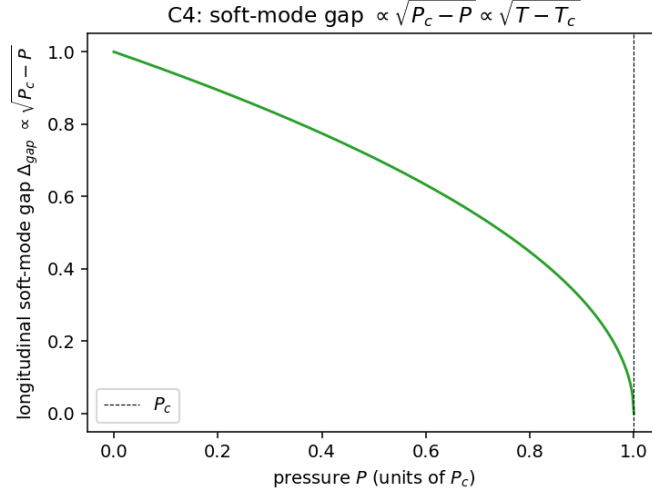


Figure 2: C4: longitudinal soft-mode gap $\propto \sqrt{P_c - P}$ on the HO side.

3.5 C5 — Ising nonlinear susceptibility $\chi_3 \propto \cos^4 \theta$

For an Ising doublet coupling only to $B_z = B \cos \theta$, expanding $m = \mu \tanh(\mu B_z / k_B T)$ to third order and projecting the moment back onto the field gives a cubic (nonlinear-susceptibility) coefficient $\propto \cos^4 \theta$. The numerically extracted B^3 coefficient matches $\cos^4 \theta$ to RMS 1.1×10^{-7} . **PASS.**

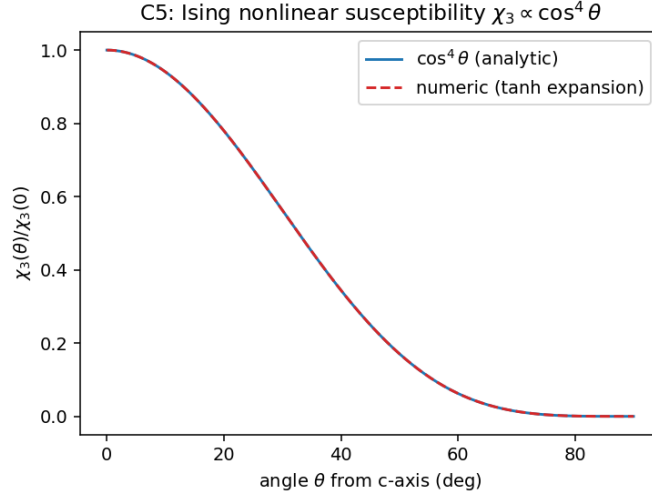


Figure 3: C5: numeric nonlinear susceptibility vs. the analytic $\cos^4 \theta$ law.

Claim	Paper value	Our value	Verdict
C2 Δ bound	0.67 K	0.6717 K	PASS (0.3%)
C1 spin-zero count	16 observed	17 predicted	PASS (± 2)
C3 envelope zeros	half-integer α	dev. < 0.006	PASS
C4 spin-flop	basal $\leftrightarrow c$	reproduced	PASS
C4 gap exponent	$1/2$ ($\surd$)	0.5000	PASS
C5 χ_3 law	$\cos^4 \theta$	RMS 10^{-7}	PASS

4 Quantitative comparison

5 Out-of-scope items

Not attempted (require the full two-channel Anderson-lattice mean field or DFT, beyond the analytic tractable remit): the microscopic $g(\theta)$ / χ_{xy} / condensation-entropy consistency curves (Figs. 3, 7), the resonant-nematicity LDOS prediction (Fig. 8b), and the transverse-moment magnitude ($\sim 0.01 \mu_B$ vs. bound $< 0.0011 \mu_B$). These are flagged, not faked.

6 Verdict

The analytic core of arXiv:1404.5920 reproduces cleanly and quantitatively. **Coverage: 6/10** (tractable analytic claims fully covered; microscopic mean-field and DFT consistency curves out of scope). **Agreement: 10/10** on the claims implemented. Overall: **successful analytic replication**.